Version 10.61->11.00 6/18/2025:

- Version number changed to 11.00 to signify major differences in firmware code (even if most functionality remained similar).
- Timekeeping change where DSP counter is used for labeling the detections time (main MCU still writes and assigns but uses value from DSP).
- Faster startup routine/sync up.
- Removed GPS syncing (sorry 3D studies for the moment)
- Removed most LED state changes
- Some small row in csv file changes
- Realtime logging

Version 11.00->11.05 7/16/2025:

- GPS fixes in SD card reimplemented, allows User control without needing specific firmware
- More GPS changes and workings to try streamlining GPS functionality
- Add USB commands similar to direct connection commands (RS232)
- Started reimplementing sensor functionality, temperature tested
- Fixed null interval bug that created time jumps (thought it was fixed)
- Changed serial output to go through either USB or RS232 if both are connected it will prefer USB
- Some receiver output cleanups
- Changed fileDump/type command. Unsure of functionality since it seems untrustworthy
- Updated dtag command
- Updated command list
- Reset command for the main supervisory chip.

Version 11.05->11.06 7/28/2025:

- More work on fileDump command since it wasn't calling the correct file(?)

Version 11.06->11.07 8/1/2025:

- Added two commands to work with fileDump called getfilename and sendfile. (removed later)
- Removed saving a fake detection during periods where there are no detections since they caused timing issues.
- Fixed hourly and daily file creation or at least known bugs at this time. Not happy with previous implementation which has now been updated but can't fix it without starting over.
- Fixed voltage value to be accurate

Version 11.07->11.08 8/5/2025:

- Changes to the firmware were made to accommodate the GPS modem labeled PA1616D which are present on newer receivers (~2022-2025). Specifically turning off
- the other GPS sentences so that just the RMC sentence is coming across (should be every second).
- Additionally, slight change in saving the GPS fix to the data, should be clean and shouldn't garble the file on the SD card.

Version 11.08->11.09 8/13/2025:

- Changes to sensors, making sure that the tilt, pressure, and temperature sensors work. This issue had to do with the sensors being a different baud rate than expected, added a check to hopefully set the correct baud rate. Tested and confirmed, however, power consumption was not tested.
- In accordance with the sensor changes, added two commands for adjusting the pressure and temperature offsets (mainly for production but customers could use).
- Added a way for sensors to send their sensor values whenever the sensors are measured, 'DisplaySensorVals' is the command (sensors have to be turned on).
- Added a way to change the rate at which the sensor measurements are taken. All sensors are tied to the same rate. 'SetSensorRate SS or -SS' sets the rate in seconds. RequestConfig sends this value out.

Version 11.09->11.10 8/20/2025:

- Changed 'settings' command. 'Settings' command should also output sensor information as well.
- Changed logic that determined what to do when detections are received. This was because the receiver firmware was using more current due to logging detections more frequently (real-time). This change should help the receiver consume less power on average. Additionally, an added benefit to this change is handling spikes of detections/noise.
- Tilt fix: Small issue was found where new receiver boards didn't give the tilt boards enough time to get an actual reading.
- Added first time checks for calibration offsets. Now when a receiver is flashed with new firmware moving from either 10.62 or 6.62 and below to 11.10 and above the offsets are maintained.

Version 11.10->11.11->11.12 9/15/2025:

- Added setting where the USB string is also set when the serial number is set. Software isn't aware of this change yet.
- Fixed issue with setting calibration offsets.
- Fixed bug where settings command was turning off GPS sentences.
- The power consumption issue was rectified for SR3001s (this was due to multiple reasons, main MCU sleep, tilt sleep, SD card writing).
- Added minimal protection against turning on tilt without a tilt board.

Version 11.12->11.13 9/17/2025:

- Reimplementing Reed Switch for SR3001 units. Changed LED patterns. Check Triton_LED_Info_v1.13. Essentially, flashing red led will mean that the hydrophone tip could be broken.

Version 11.13->11.14 10/15/2025:

- Changed LEDs for 3001s. Making it so that LEDs will be on for the first 5 minutes and then will only be turned on when a magnet is swiped (for 2 minutes).

Version 11.14->11.15 10/22/2025:

- 1 GB file size limit implemented. If a file size goes above 1 GB a new log file should be created on the SD card. This is due to the FAT32 architecture.

Version 11.15->11.16 10/29/2025:

- Added acknowledgement print statements after a command is sent
- Added setGPSfreq value to settings command output
- Added calibration values for tilt boards
- Added output statements for tilt calibration values

Version 11.16->11.17 1/26/2026-3/4/2026:

- Added “type” command as an alias for the command “fileDump”
- Fixed missing hourly character for hourly file creation (intermittent issue before)
- Additionally, seconds are now back in the file name header instead of file count (last 2 digits)
- Tilt fix, issues were still appearing where every 1 in 10 receivers was showing issues with reading/displaying tilt values. This tilt fix addresses this issue and ideally removes the need for changing the tilt calibration constants that were added in 11.09/11.10. An added benefit to this fix is keeping the receiver more responsive since the previous tilt method was slower (couple of seconds)
- The tilt fix issue resolution caused pressure to have some trouble, updated code to account for this.
- Fixed an erroneous linear addition to time on ~11-day rollover time event. This would only occur if multiple detections occurred right at the rollover event.

Version 11.17->11.18 4/8/2026-5/11/2026:

- Requesting a file from the receiver that is not correctly the active file can potentially/commonly corrupt the file that is requested. Additionally, after the file is sent the receiver does not resume logging from the previous file. As far as is known, this is present in all firmwares. Fix implementation: Stores the active log file directory information so the receiver can correctly return to logging after dumping a request file.
- When working with hourly or daily file creation some receivers have corrupted SD cards. It is believed that this is due to the receiver having an error in the file creation logic or something internal. Exact cause is unknown, however, this issue is being investigated. From what is gathered so far, the receiver may mishandle the FAT directory and this causes the corruption. Fix: It is thought that the FAT allocation is being done wrong, allocating from the ROOT index instead of the index after the root. Testing is difficult since the issue isn’t consistent or easily reproducible.
- Receiver has issues when it gets no detections for over a day making the time not increment forward as expected. Fix: Day rollover logic wasn’t accounted for in incrementing the timestamp.
- GPS fixes were arriving at the same time that tag detections were arriving, so they were colliding with detection data. This created race conditions that sometimes would result in corrupted lines. Fix: Storing GPS fix times in a buffer when they arrive allows us to delay writing the GPS fixes to the card and helps us prevent collisions. Previously, the GPS fixes would store the data similarly to tag detection as this was the previous setup by Nancy.
- GPS time syncing using RS232 command 5MsyncTog (can’t toggle via USB, must be RS232). Time period is set to “10 minutes”. Though initial testing it varied and seemed to happen every 20 or 40 minutes. Still verifying time of the sync. May implement fix to SD

card to notify SD card of this event. Though it can be sniffed out by looking at the internal column.

Version 11.18->11.19 5/21/2026:

- Debug/serial messages cleanup. The function in the firmware that sends messages out the RS232/USB port was implemented in a resource intensive way, cleaning this up should free up resources and protect against receiver overrun time. Additionally, there were some ISR calls that seemed like they could potentially cause bugs, so these were addressed.
- Removed USB commands for older software since they do not seem to work and the new software should be used instead.
- Added 5MsyncTog to USB commands
- GPS syncs are now recorded on the SD card with “PREV TIME” “NEW TIME”. There will always be at **least** a two second difference since the process takes 2 seconds. NewTime – Prevtime – 2 = “Supposed Drift”. There is currently no way to measure DSP drift smaller than one second (without hardware changes). Therefore these 5 minute sync, if they happen consistently, may not “show” drift occuring.
- GPS fixes in the SD card will have a count in the signal strength column that will display the number of consecutive GPS sentences received from the GPS modem. This is helpful since the GPS time syncs are only done when 5 minutes have passed and there have been 10 or more consecutive GPS sentences. Additionally, these GPS time syncs take “2” seconds of offline to occur (DSP is not decoding for 1 second).
- Updated command list

Version 11.19->11.20 6/3/2026:

- Lines of nul characters no longer present (they were not always displayed). This was occurring because of incorrect file structure management.

Version 11.20->11.21 6/24/2026 (testing):

- A setting that can be set on a receiver to write time sync times to the SD card. The default is that it is ON. ‘logsync’ command is the appropriate toggle command to turn it off.
- Some USB commands “fall” through to a different command. This was missed because the legacy software had multiple single letter commands that did the same thing. Corrected so they don’t fall through (may be used in the future for something else).
- Changed how the receiver acknowledges USB commands to save on memory. It should now first acknowledge a command was received and echo back the letter received and then proceed to the command actions.

Future Update list:

- Possibly make GPS time syncing period user set like how GPS fixes to the SD card are also user set (currently it is 5 minute check).
- Possibly add nulltags back in(?)